



COLLEGE OF
COMPUTER STUDIES

Week 4

FOPR111 – Fundamentals of Programming

Algorithms, Pseudocode Introduction to Flowchart

Objectives:

By the end of this lesson, you should be able to:

- Define what is Algorithm and Pseudocodes
- Know the uses of Algorithms and Pseudocodes
- Knowledge in using the different Flowchart symbol and their functions.
- Illustrate a problem analysis through flowcharting.
- Produce different kinds of problem-solving solutions using the concepts of flow charts



Algorithms

Algorithm

- A set of step-by-step instructions to accomplish a task.
 - An algorithm must have start instruction
 - Each instruction must be precise.
 - Each instruction must be unambiguous.
 - Each instruction must be executed in finite time.
 - An algorithm must have stop instruction.

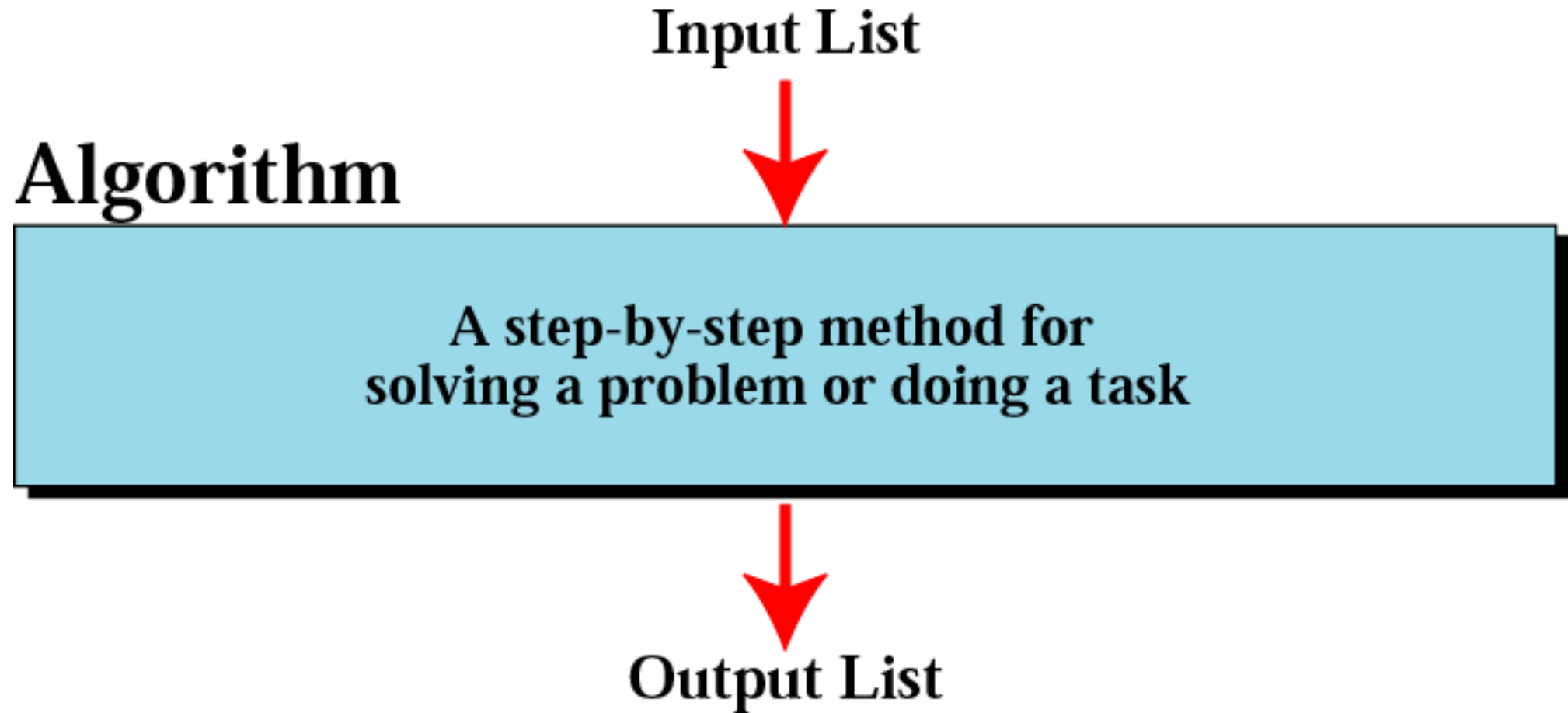
Algorithm

- originally referred to any computation performed via a set of rules applied to numbers written in decimal form.
- word is derived from the phonetic pronunciation of the last name of Abu Ja'far Mohammed ibn Musa al- Khowarizmi, who was an Arabic mathematician who invented a set of rules for performing the four basic arithmetic operations

Algorithm

- A procedure is a finite sequence of well-defined instructions, each of which can be mechanically carried out in a finite amount of time.

Basic Algorithm Concept



Algorithm to find the sum of two number

Step 1 : Start

Step 2 : Declare variable num1 , num2 and sum

Step 3 : Read values num1 and num2

Step 4 : Calculate the sum of num1 and num2

Step 5 : Display sum

Step 6 : Stop

Input = Enter two numbers (num1, num2)

Process = Compute the sum ($\text{sum} = \text{num1} + \text{num2}$)

Output = Display the sum (sum)

Algorithm to find the area of a rectangle

Step 1 : Start

Step 2 : Declare variable length, width, areaR

Step 3 : Read values length and width

Step 4 : Multiply length and width and assign the result to areaR

Calculate the areaR using the formula $\text{length} * \text{width}$

Step 5 : Display areaR

Step 6 : Stop

Input = Enter length and width (length, width)

Process = Compute the area ($\text{areaR} = \text{length} * \text{width}$)

Output = Display the area (areaR)

Algorithm to find the largest of two numbers

Step 1 : Start

Step 2 : Declare variable val1, val2 and ans

Step 3 : Read values val1 and val2, compare val1 and val2

Step 4 : If $A > B$ then print "A is largest" and go to step 6

Step 5 : "B is largest"

Step 6 : Stop

Input = Enter val1 and val2 (val1, val2)

Output = Display the answer (ans)

Process = if (val1 > val2)

 print ("val1 is largest");

 else

 print ("val2 is largest");



Flowchart

Flowchart

- is a tool developed in the computer industry, for showing the steps involved in a process.
- A flowchart is a diagram made up of boxes, diamonds and other shapes, connected by arrows - each shape represents a step in the process, and the arrows show the order in which they occur.

Flowchart

- A flowchart is a diagram that depicts the “flow” of a program.

Flowcharting Basic Symbols



Start/End Symbol

The terminator symbol marks the starting or ending point of the system. It usually contains the word "Start" or "End."



Action or Process Symbol

A box can represent a single step ("add two cups of flour"), or an entire sub-process ("make bread") within a larger process.



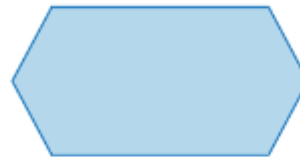
Input/Output Symbol

Represents material or information entering or leaving the system, such as customer order (input) or a product (output).



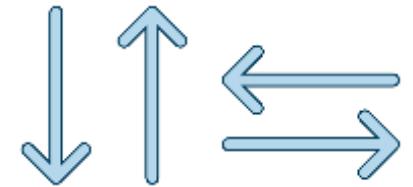
Decision Symbol

A decision or branching point. Lines representing different decisions emerge from different points of the diamond.



Preparation Symbol

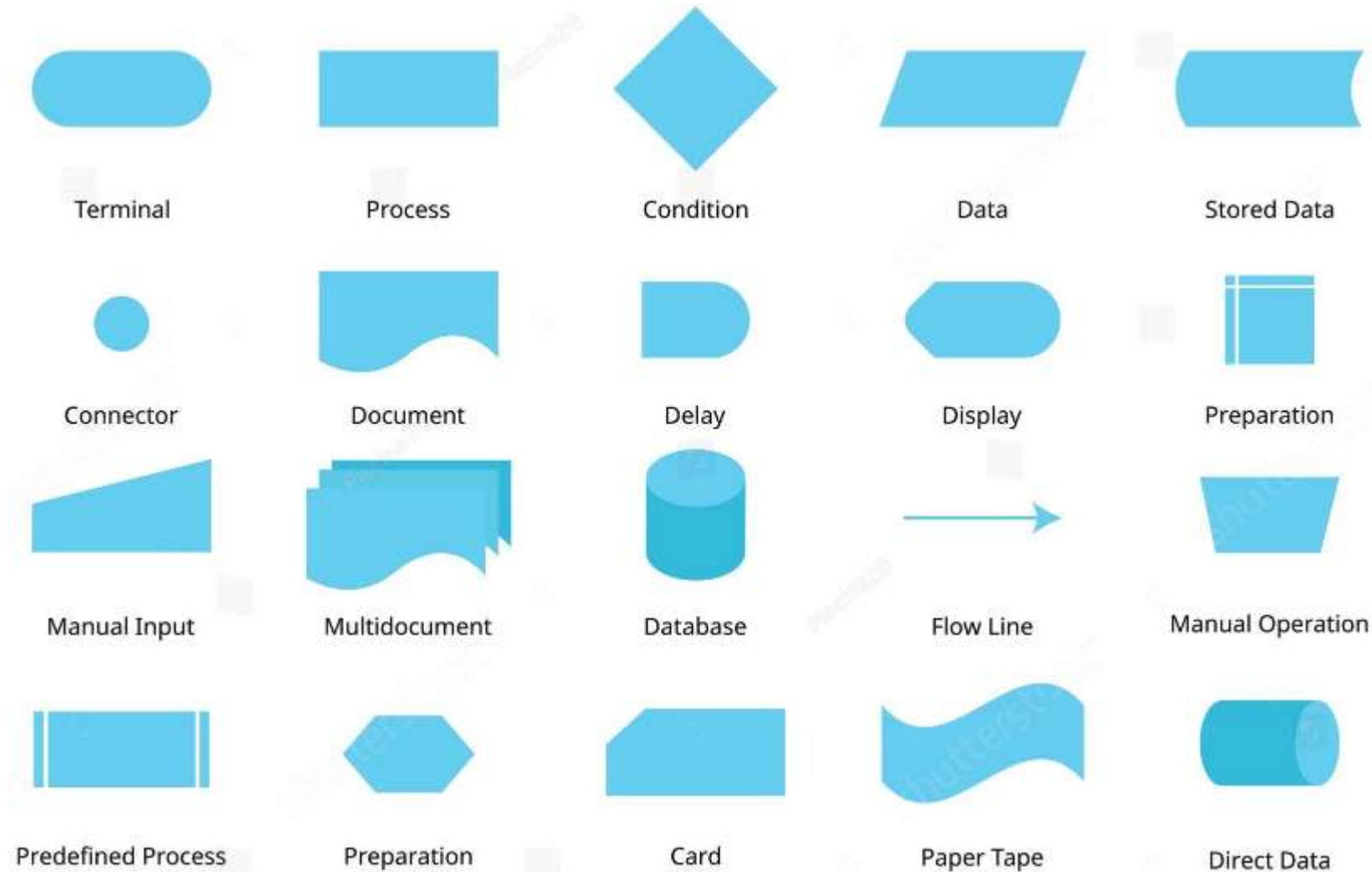
Represents a set-up to another step in the process.



Flow Lines

Show direction of flow.

Flowcharting Basic Symbols



General Rules for flowcharting

All boxes of the flowchart are connected with Arrows. (Not lines)

Flowchart symbols have an entry point on the top of the symbol with no other entry points. The exit point for all flowchart symbols is on the bottom except for the Decision symbol.

The Decision symbol has two exit points; these can be on the sides or the bottom and one side.

Generally, a flowchart will flow from top to bottom. However, an upward flow can be shown as long as it does not exceed 3 symbols.

Connectors are used to connect breaks in the flowchart.

Subroutines and Interrupt programs have their own and independent flowcharts.

All flow charts start with a Terminal or Predefined Process (for interrupt programs or subroutines) symbol.

All flowcharts end with a terminal.

Algorithm/Flowchart to find the sum of two numbers

Step 1 : Start

Step 2 : Declare variable num1 , num2 and sum

Step 3 : Read values num1 and num2

Step 4 : Calculate the sum of num1 and num2

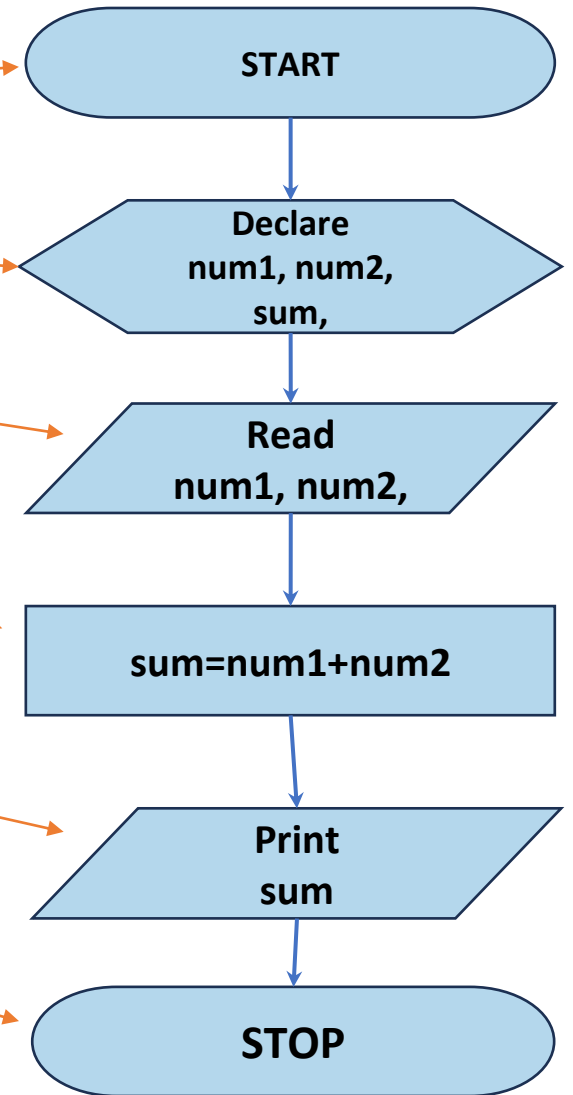
Step 5 : Display sum

Step 6 : Stop

Input = Enter two numbers (num1, num2)

Process = Compute the sum ($\text{sum} = \text{num1} + \text{num2}$)

Output = Display the sum (sum)



Algorithm to find the largest of two numbers

Step 1 : Start

Step 2 : Declare variable val1, val2 and ans

Step 3 : Read values val1 and val2, compare val1 and val2

Step 4 : If $A > B$ then print "A is largest" and go to step 6

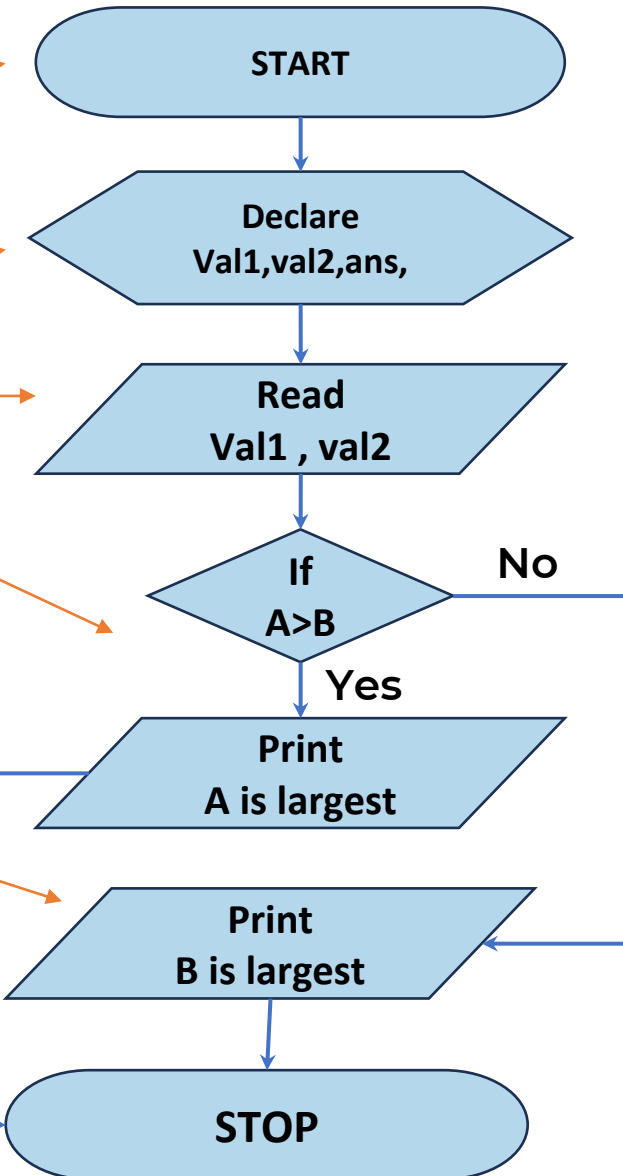
Step 5 : "B is largest"

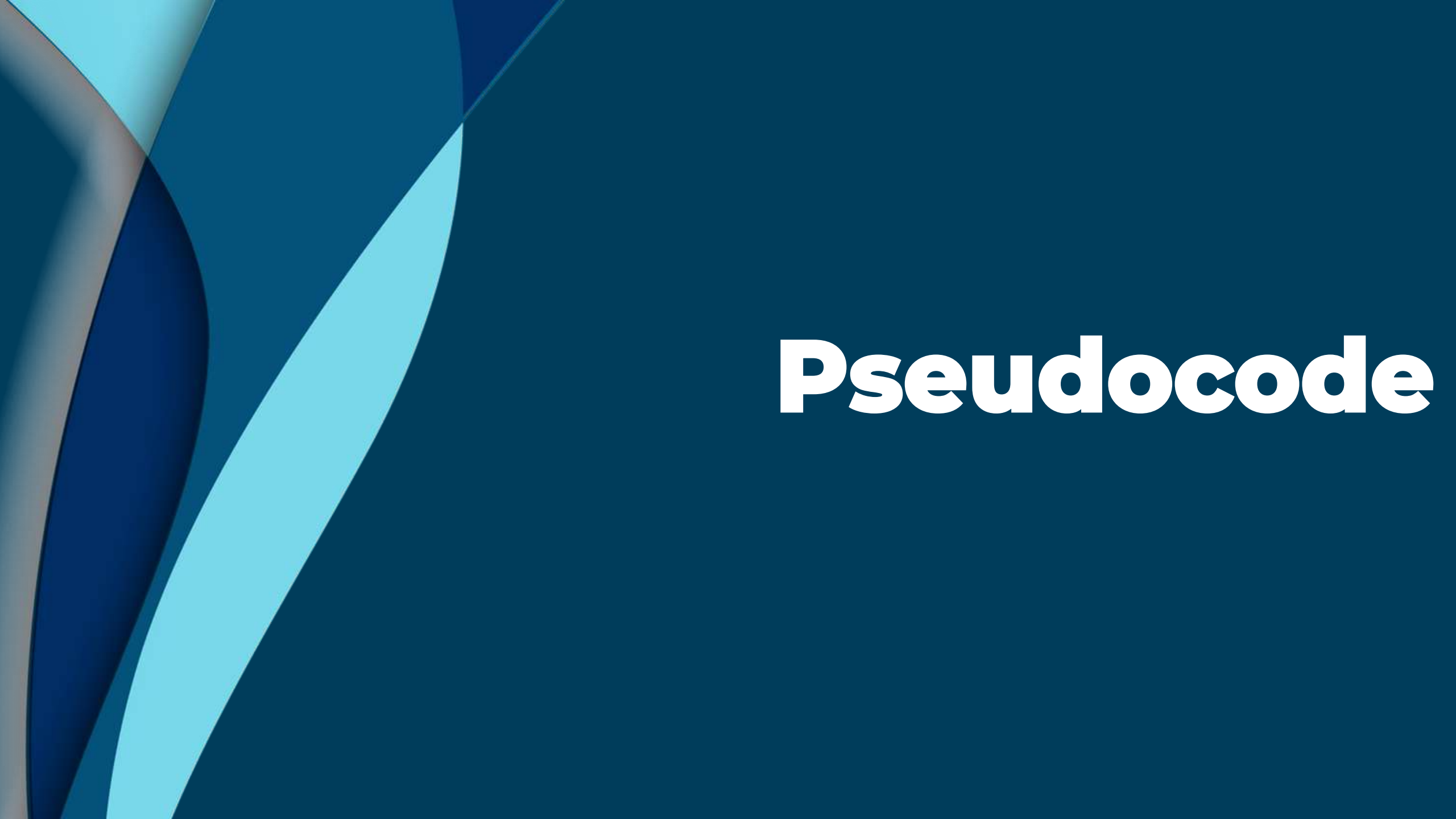
Step 6 : Stop

Input = Enter val1 and val2 (val1, val2)

Process = if ($A > B$)
 print ("A is largest");
else
 print ("B is largest");

Output = Display the answer (ans)





Pseudocode

Pseudocode

- is an artificial and informal language that helps develop algorithms.
- It is useful in developing algorithms that will be later converted into C language programming.
- It is similar to English statements and not executed on computers.

Pseudocode Constructs

These constructs — also called keywords — are used to describe the control flow of the algorithm.

- **SEQUENCE** represents linear tasks sequentially performed one after the other.
- **WHILE** is a loop with a condition at its beginning.
- **REPEAT-UNTIL** is a loop with a condition at the bottom.
- **FOR** is another way of looping.
- **IF-THEN-ELSE** is a conditional statement changing the flow of the algorithm.
- **CASE** is the generalization form of IF-THEN-ELSE.

Pseudocode Constructs

SEQUENCE

Input: READ, OBTAIN, GET
Output: PRINT, DISPLAY, SHOW
Compute: COMPUTE,
CALCULATE, DETERMINE
Initialize: SET, INIT
Add: INCREMENT, BUMP
Sub: DECREMENT

FOR

FOR iteration bounds
sequence
ENDFOR

WHILE

WHILE condition
sequence
ENDWHILE

CASE

CASE expression OF
condition 1: sequence 1
condition 2: sequence 2
...
condition n: sequence n
OTHERS:
default sequence
ENDCASE

REPEAT-UNTIL

REPEAT
sequence
UNTIL condition

IF-THEN-ELSE

IF condition THEN
sequence 1
ELSE
sequence 2
ENDIF



Algorithm/Pseudocode to find the area of a rectangle

Step 1 : Start

Step 2 : Declare variable length, width, areaR

Step 3 : Read values length and width

Step 4 : Calculate the areaR using the formula $\text{length} * \text{width}$

Step 5 : Display areaR

Step 6 : Stop

Start

Set length=0, width=0, areaR=0;

Read length,width;

Compute areaR=length * width;

Print areaR;

Stop

Input = Enter length and width (length, width)

Process = Compute the area (areaR =length+width)

Output = Display the area (areaR)

Algorithm /Pseudocode to find the largest of two numbers

Step 1 : Start

Step 2 : Declare variable val1, val2 and ans

Step 3 : Read values val1 and val2

Step 4 : If A>B then print "A is largest" and go to step 6

Step 5 : "B is largest"

Step 6 : Stop

Input = Enter val1 and val2 (val1, val2)

Process = if (A>B)

 print ("A is largest");

 else

 print ("B is largest");

Output = Display the answer (ans)

Start

INIT val1=0, val2=0, ans=0;

GET val1 ,val2

IF A>B THEN

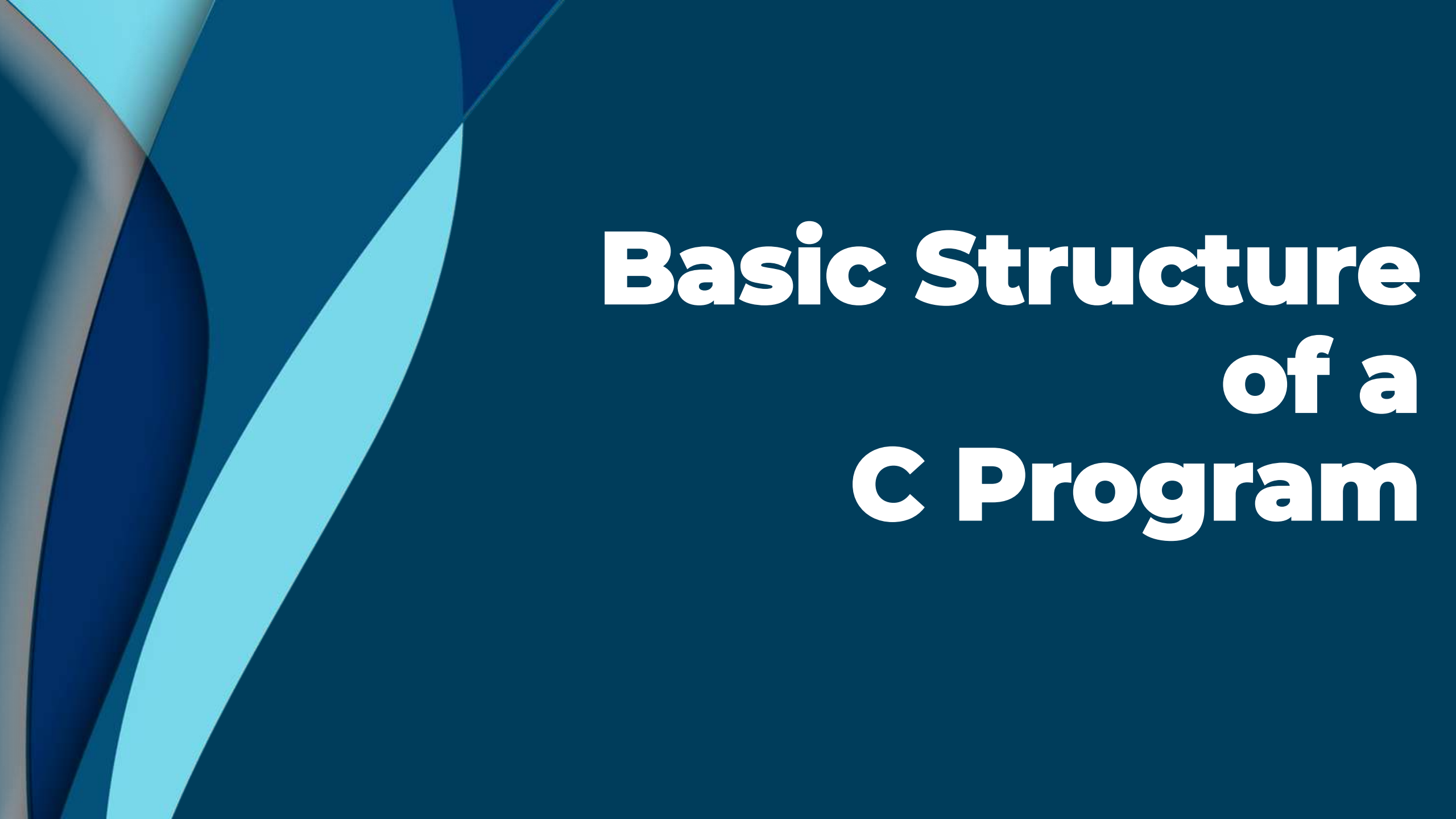
 print "A is largest"

ELSE

 print "B is largest"

ENDIF

Stop



Basic Structure of a C Program

Basic Structure of a C Program

```
#include <stdio.h>
```

```
int main() {
```

```
// your code
```

```
printf("Hello World!");
```

```
return 0;
```

```
}
```

`#include <stdio.h>` is a **header file library** that lets us work with input and output functions, such as `printf()` (used in line 4). Header files add functionality to C programs.

`main()`. This is called a **function**. Any code inside its curly brackets `{ }` will be executed.

`printf()` is a **function** used to output/print text to the screen

`return 0` ends the `main()` function.

Note that: Every C statement ends with a semicolon ;

Thank you

Let's do something great

